# Student Success Through Artificial Intelligence

## (Machine Learning–Based Warning System for Student Dropout Risk)

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Scope of Presentation



# Why This Project Matters

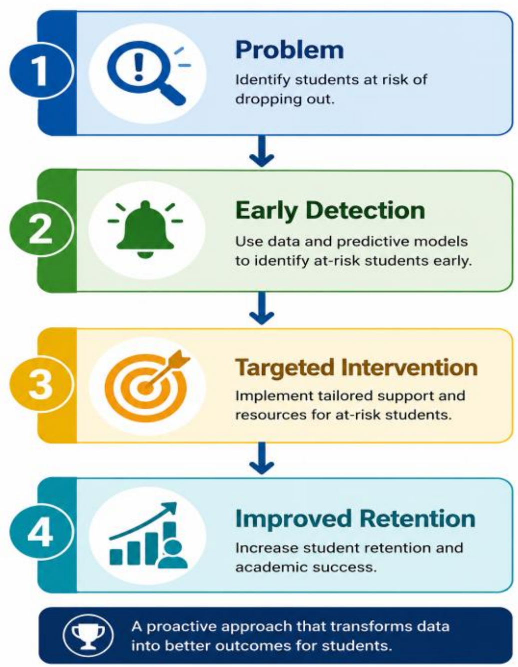
## The Challenge

Student dropout has significant academic and institutional consequences. Impacts include

- Lower graduation rates
- Reduced student success
- Inefficient allocation of institutional resources
- Missed opportunities for timely intervention

## Our Objective

Develop an AI-powered predictive model that identifies students who are likely to drop out before they leave the institution.

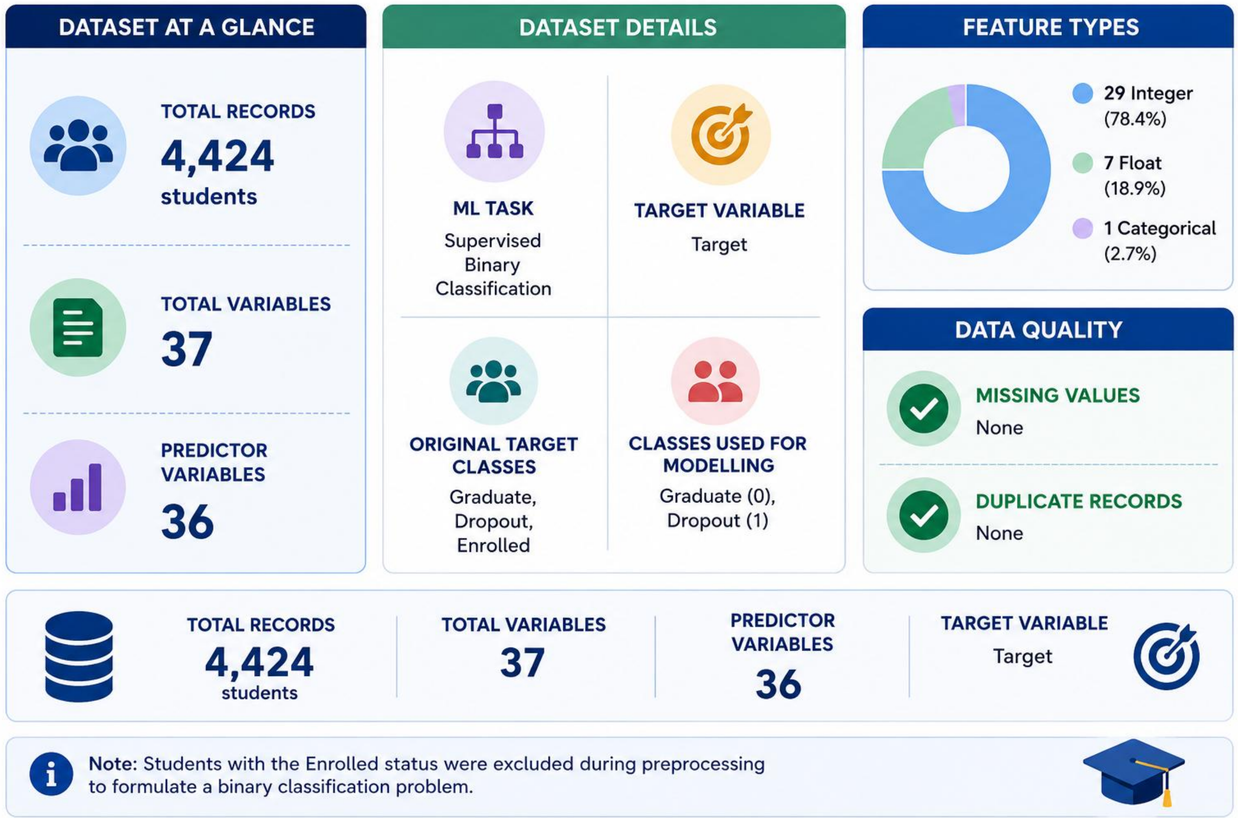



. Early identification of at-risk students allows the institution to provide timely academic advising, mentoring, counselling, and financial support.

Scope of Presentation



# Key Dataset Facts

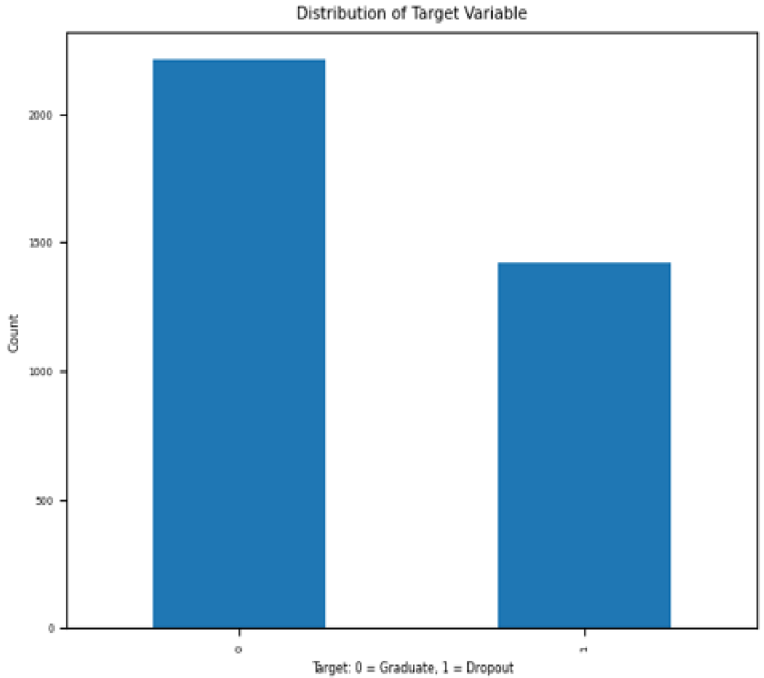


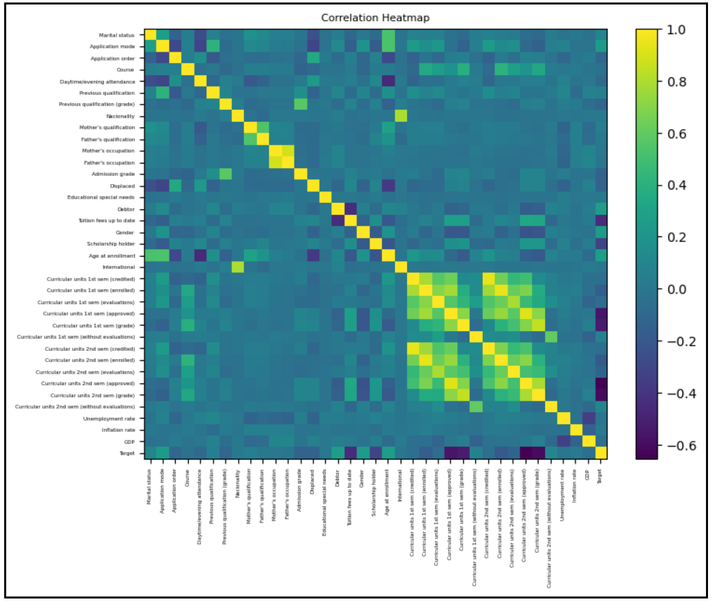

**Executive Insight**: The dataset provides a comprehensive view of students' academic, demographic, financial, and enrollment characteristics. This breadth of information enabled the development of a robust machine learning model for identifying students who may be at risk of dropping out.

## Scope of Presentation

- Why the Project Matters
- Key Dataset Facts
- Key Insights from Data
- AI Solution Overview
- Key Factors Associated with Dropout
- Risks & Considerations
- Business and Educational Value
- Results and Model Comparison
- Recommendations

# Key Insights from Data




Exploratory data analysis confirmed that the dataset contains meaningful patterns suitable for predictive modelling. The target distribution and correlation analysis provided the foundation for feature engineering and model development

## Scope of Presentation



# AI Solution Overview


The project follows a structured artificial intelligence workflow that transforms institutional student data into actionable insights for early dropout prediction. It begins with data cleaning and exploratory data analysis, followed by feature engineering, feature selection, and dimensionality reduction to prepare the data for modeling. Multiple machine learning models are then trained, evaluated, and optimized to identify the best-performing model, which is saved for future use. The resulting AI solution serves as a decision-support tool that helps the university identify at-risk students and support timely, data-informed interventions to improve student retention and academic success

# Key Factors Associated with Dropout

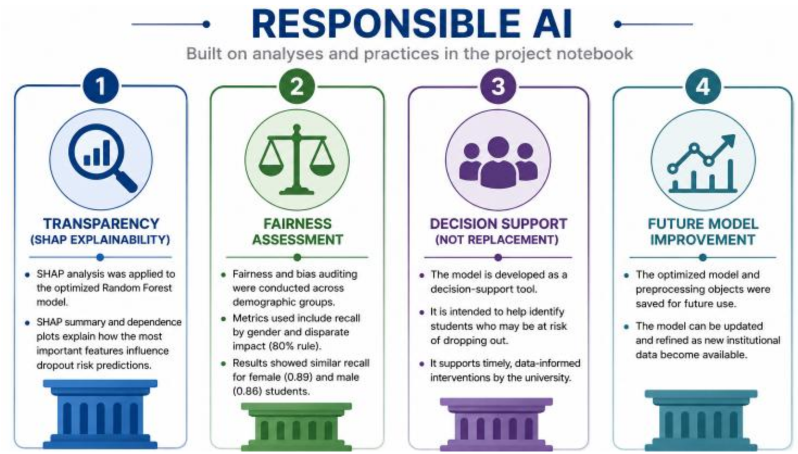
| | Feature | Importance |
|---|---|---|
| 36 | Total approved units | 0.166784 |
| 30 | Curricular units 2nd sem (approved) | 0.156072 |
| 24 | Curricular units 1st sem (approved) | 0.091018 |
| 31 | Curricular units 2nd sem (grade) | 0.077530 |
| 37 | Average semester grade | 0.071256 |
| 25 | Curricular units 1st sem (grade) | 0.045186 |
| 16 | Tuition fees up to date | 0.044036 |
| 12 | Admission grade | 0.026141 |
| 19 | Age at enrollment | 0.024359 |
| 29 | Curricular units 2nd sem (evaluations) | 0.023649 |
| 3 | Course | 0.023384 |
| 6 | Previous qualification (grade) | 0.021573 |
| 22 | Curricular units 1st sem (enrolled) | 0.020123 |
| 23 | Curricular units 1st sem (evaluations) | 0.018950 |
| 28 | Curricular units 2nd sem (enrolled) | 0.018416 |

The model identified students' academic progress and performance as the most influential predictors of dropout risk, providing evidence to support earlier and more targeted intervention strategies

# Risks and Considerations

## Ensuring Responsible AI Adoption

To promote responsible AI, the project integrates explainability and fairness assessments. SHAP explains the key factors behind predictions, while fairness analysis ensures consistent performance across demographic groups, supporting transparent and ethical decision-making




### Key Considerations:

- ✓ AI supports—not replaces—human decision-making.
- ✓ Predictions should guide timely academic interventions.
- ✓ Model performance should be periodically monitored using updated institutional data.

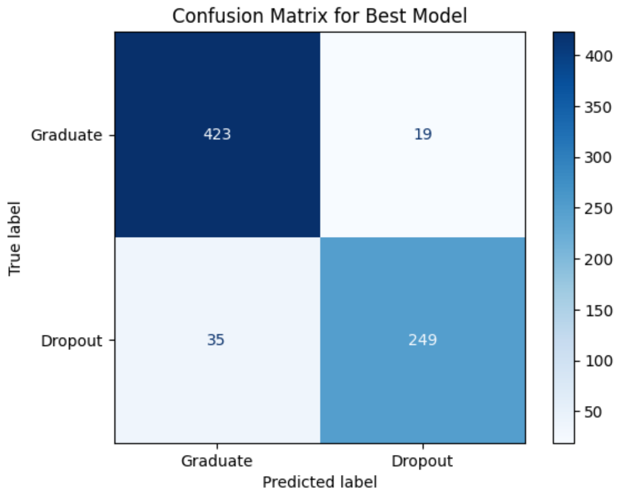
# Business and Educational Value

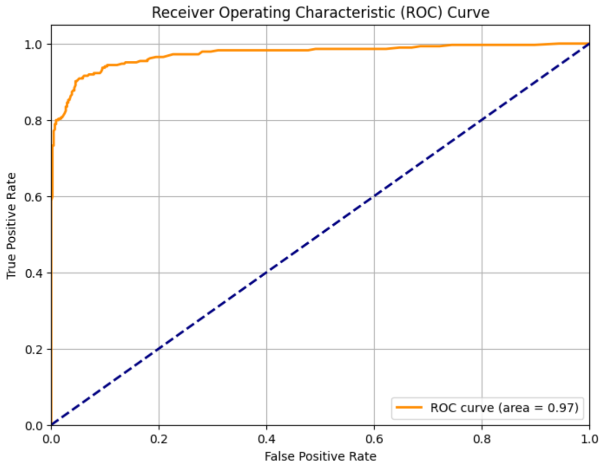
The optimized AI model provides university leaders with a decision-support tool for identifying students who may be at risk of dropping out. By enabling earlier identification of at-risk students, the model can support timely academic interventions, support more informed allocation of institutional resources, and strengthen evidence-based student support initiatives


**Expected Institutional Benefits**

- **Early Identification** – Detect students who may require academic support sooner.
- **Targeted Advising** – Help prioritize advising, mentoring, counselling, and other student support services.
- **Improved Student Success** – Support initiatives that enhance student retention and academic achievement.
- **Better Resource Allocation** – Enable more informed allocation of institutional resources to students who need them most.

# Results and Model Comparison

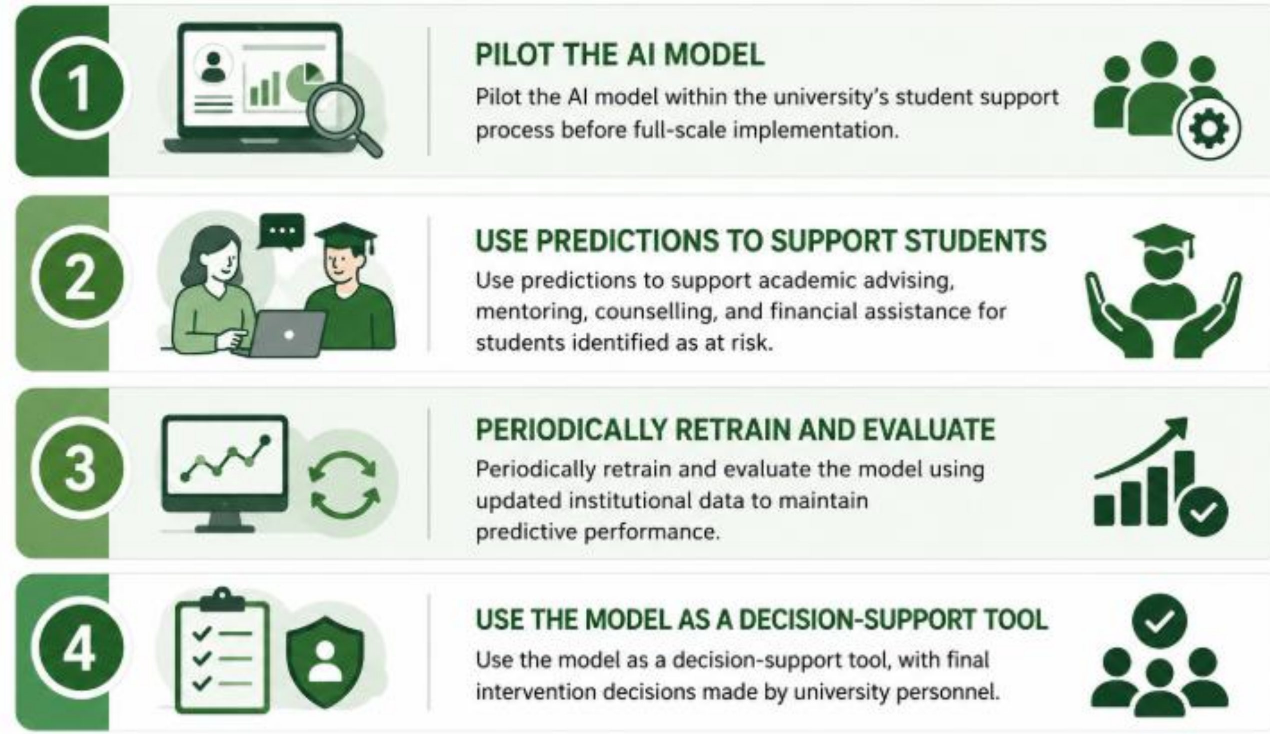



| Evaluation Stage | Outcome |
|---|---|
| Multiple predictive models tested | Four machine learning models were evaluated. |
| Performance comparison | Models were assessed using standard accuracy and reliability measures. |
| Best-performing solution | Random Forest demonstrated the strongest overall predictive performance. |
| Model optimization | The selected model was further improved using automated parameter tuning. |
| Final model | Optimized Random Forest selected for deployment. |

# Recommendations

## Four-step Implementation Roadmap




Implementing these recommendations can help the university adopt AI responsibly while strengthening evidence-based student support and retention initiatives

**End of Presentation**

**Good Day to Everyone**